



impact while laying the foundation for more advanced innovations in the future.

Role of OEMs: Owning the architecture

A critical missing link in the current ecosystem is the limited involvement of OEMs, particularly in sectors like automotive. Traditionally, many OEMs rely heavily on externally developed architectures and pre-integrated solutions, which restrict their ability to differentiate, optimise, or innovate at a system level.

To unlock the full potential of indigenous development, OEMs must take on a more proactive role by owning the system architecture. This begins with clearly defining:

- The functional requirements of the system
- The specific use cases relevant to Indian conditions
- Performance, safety, and cost constraints

Once this architectural blueprint is established, OEMs can collaborate with startups, semiconductor companies, and research institutions to develop tailored solutions.

Such an approach enables a modular and collaborative ecosystem in which multiple players contribute specialised components that integrate into a cohesive system.

The benefits of this shift are significant. It enables the creation of customised solutions optimised for Indian environments, improves safety and performance through tighter control, and fosters scalable innovation by opening opportunities for smaller players to participate in the value chain.

Building a collaborative ecosystem

For India to transition from aspiration to execution, it must adopt a collaborative, ecosystem-driven approach. No single entity—whether government, industry, or academia—can achieve this transformation in isolation.

First and foremost, there is a need to break down silos and encourage collaboration. Startups, academic institutions, OEMs, and government bodies must work in tandem, sharing knowledge, resources, and infrastructure. A cooperative model will not only reduce duplication of effort but also accelerate innovation and standardisation.

Equally important is the development of indigenous tools, particularly in the domain of EDA and simulation platforms. These tools form the backbone of semiconductor design, and without local alternatives, true independence remains elusive.

Another critical area is the creation of shared prototyping infrastructure. Accessible labs, testing facilities, and compliance-ready environments can empower students and early-stage innovators to transform ideas into working prototypes, bridging the gap between theory and practice.

In parallel, India must place a strong emphasis on excellence in validation. Testing and verification should not be treated as an afterthought but integrated into the earliest stages of design. This ensures that products are robust, reliable, and ready for real-world deployment.

Finally, the ecosystem must invest in developing hybrid talent: engineers who are equally comfortable working with hardware, firmware, and software. Such multidisciplinary expertise is essential for building complex, integrated systems.

The concept of indigenisation is often misunderstood as simply replacing imports with locally manufactured alternatives. However, this is a narrow and incomplete view. True indigenisation is about ownership and control. It means having the ability to:

- Understand and govern system behaviour
- Diagnose and fix issues independently

• Customise solutions for specific needs

• Ensure security, reliability, and long-term sustainability

In this context, self-reliance is not just about where a product is made; it is about who controls its intelligence and decision-making.

Building a robust and indigenous semiconductor ecosystem is inherently a long-term endeavour. It requires sustained commitment, significant investment, and a willingness to navigate long development cycles. Unlike software, where results can be realised quickly, semiconductor innovation demands patient capital and strategic foresight.

Equally important is the need for ecosystem-wide alignment. Success will depend on how effectively different stakeholders, government, industry, academia, and startups, align their efforts towards a common goal.

The rewards of this journey, however, are profound. Achieving technological sovereignty will not only strengthen national security but also drive economic growth, create high-value jobs, and position India as a competitive force in the global technology landscape.

Today, India stands at a pivotal moment. The choices made now will determine whether the country remains a consumer of advanced technologies or evolves into a creator and controller of its own technological future.

The semiconductor journey is not about catching up with the world, it is about building with clarity, conviction, and control. With the right alignment, India will not merely participate in the global electronics ecosystem, it will help define its next chapter. **EFY**

This article is based on a technical session delivered at Pune Expo 2026, Pune, by Poonam Gupta, Founder/Innovator, VayuLink Semiconductor, Swaraj Silicon/IndiChip Founder Circle. It was transcribed and curated by Akanksha Sondhi Gaur, Senior Technical Journalist at EFY.